Exact enumeration of five-row triangular arrays with equal corresponding line sums

GPT-6 Astra

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Abstract

We prove the six constituent formulas conjectured for OEIS A195806. The arrays have entries in $\{0,\dots,n\}$, zero corners, and equal sums on the three lines of each fixed length. An integral six-parameter bijection reduces the count to a sum of products of two squared interval lengths. A dihedral symmetry of the two difference parameters gives an elementary rational generating function and all six formulas, for every $n \geq 0$. The quasipolynomial has degree six and minimal period six. An independent polytope argument proves the period bound using only fifteen two-by-two determinants. We include exact arithmetic checks, boundary multiplicities, and an explicit correction of a source-index discrepancy and a printed coefficient error in the 2023 conjecture.

1 The counting problem and its resolution

Put

$$I = \{(i,j,k) \in \mathbb{Z}^3_{\geq 0} : i+j+k = 4\}.$$

An admissible array is a map $u : I \to \{0,\dots,n\}$ with $u_{400} = u_{040} = u_{004} = 0$, such that, for $c = 0,1,2,3,4$,

$$\sum_{i=c} u_{ijk} = \sum_{j=c} u_{ijk} = \sum_{k=c} u_{ijk}. \tag{1}$$

Each sum runs over I. Its line length is $5-c$. There is no equality condition between lines of different lengths. All positions are labeled: arrays related by rotations or reflections are counted separately. Write $a(n)$ for their number; in particular $a(0) = 1$.

Theorem 1. *For every integer $m \geq 0$ and $0 \leq r < 6$,*

$$a(6m+r) = \sum_{j=0}^{6} c_{r,j} m^j,$$

where the coefficients are exactly those in Table 1. The degree is six and the least quasiperiod is six.

The $r = 5$ constituent is equivalently the factorization

$$a(6m+5) = (m+1)^2(5052 + 19370m + 28405m^2 + 18720m^3 + 4680m^4). \tag{2}$$

Theorem 2. *As an identity of formal power series,*

$$F(z) := \sum_{n\geq 0} a(n) z^n = \frac{N(z)}{(1-z)^7(1+z)(1+z+z^2)^3}, \tag{3}$$

where

$$\begin{aligned} N(z) = 1 &+ 13z + 59z^2 + 212z^3 + 471z^4 + 753z^5 + 882z^6 \\ &+ 753z^7 + 471z^8 + 212z^9 + 59z^{10} + 13z^{11} + z^{12}. \end{aligned}$$

r	$c_{r,0}$	$c_{r,1}$	$c_{r,2}$	$c_{r,3}$	$c_{r,4}$	$c_{r,5}$	$c_{r,6}$
0	1	25	158	650	2275	4680	4680
1	16	198	1133	3900	8125	9360	4680
2	105	1087	4922	12350	17875	14040	4680
3	496	4148	14783	28600	31525	18720	4680
4	1759	12121	35258	55250	49075	23400	4680
5	5052	29474	72197	94900	70525	28080	4680

Table 1: The six constituent polynomials, including every boundary case.

The proof below is an all-n enumeration followed by rational algebra. The finite checks in Section 6 are additional verification. They are not used to guess a period.

Related work, attribution, and indexing

Hardin introduced the array sequence in OEIS A195806 [1]. The six formulas in Table 1 were proposed by Kauers and Koutschan. Their article [2, §5.1, pp. 24–26] states the closed form as Conjecture 11 and proves D-finiteness in Theorem 13; their discussion also establishes the existence of a quasipolynomial. Thus neither the counting problem, the formulas, nor the existence of a quasipolynomial originates here. This paper supplies an explicit proof of the six constituents.

The OEIS offset is 1, with initial entries $16, 105, 496$. Our natural bound convention gives $a(1) = 16$, $a(2) = 105$, $a(3) = 496$, and appends $a(0) = 1$. There is a separate discrepancy in the expression literally printed in Conjecture 11 of [2]. If it is denoted by $B(t)$, then exact polynomial comparison gives

$$B(n-1) - a(n) = \begin{cases} 1/48, & n \equiv 0 \pmod 6, \\ 0, & n \not\equiv 0 \pmod 6. \end{cases} \tag{4}$$

Replacing its residue-5 constant 19496 by 19469 makes the corrected expression $B(t) = a(t+1)$. The shift is necessary as well as the coefficient correction. For example the printed expression at $t = 5$ is $12469 + 1/48$, whereas $a(6) = 12469$. The six conjectured formulas proved here require no correction.

The literature audit on 28 October 2026 checked the journal version, the latest listed arXiv version (v2, 24 April 2023), the authors' computational supplement, and later-publication searches. No later proof of these exact six constituents was located. This bounded search does not establish worldwide priority.

2 An integral bijection and exact interval counts

List the twelve noncorner positions in lexicographic order:

$$013,\ 022,\ 031,\ 103,\ 112,\ 121,\ 130,\ 202,\ 211,\ 220,\ 301,\ 310. \tag{5}$$

The abbreviation 013 means $(0, 1, 3)$, and similarly throughout.

Lemma 3. *The real solutions of (1) with zero corners are precisely*

$$\begin{aligned}(a+2p+q,\ c-3p-q,\ e+p,\ e-p-q,\ f+3p+q,\ f+3p+2q,\\ a,\ c+q,\ f,\ c,\ a+p,\ e),\end{aligned} \tag{6}$$

in the order (5), with arbitrary real parameters (a, c, e, f, p, q). For integer arrays, these parameters range over exactly $\mathbb{Z}^6$.

Proof. Given a solution, set

$$a = u_{130},\ c = u_{220},\ e = u_{310},\ f = u_{211}, \quad p = u_{301} - u_{130}, \quad q = u_{202} - u_{220}. \tag{7}$$

The level-3 equations give $u_{031} = e + p$ and $u_{013} + u_{103} = a + p + e$. At level 0 the common sum is $u_{130} + u_{220} + u_{310} = a + c + e$. Hence

$$u_{103} = e - p - q, \qquad u_{013} = a + 2p + q, \qquad u_{022} = c - 3p - q.$$

The level-2 equations, whose common sum is $2c + q + f$, then give $u_{112} = f + 3p + q$ and $u_{121} = f + 3p + 2q$. This proves necessity. Conversely substitution verifies equality at levels $0, 2, 3$, and the level-4 corner sums are zero. For each direction, the sum of the five parallel line sums is the sum of all entries. Consequently equality at the other four levels implies equality at level 1 as well. Formula (7) is an integral inverse to (6); there is no extra congruence condition or lattice-index factor. □

For fixed p, q, group the entries into four triples:

base	offsets		
a	0	p	$2p + q$
c	0	q	$-3p - q$
e	0	p	$-p - q$
f	0	$3p + q$	$3p + 2q$.

(8)

If the offsets form a set S, the allowable integral base lies in the closed interval $[-\min S, n - \max S]$. Its number of choices is

$$\max(0, n + 1 - (\max S - \min S)).$$

In particular both endpoints are included. The span of three numbers is the largest absolute difference of a pair. Define

$$\begin{aligned} A(p, q) &= \max(|p|, |p + q|, |2p + q|), \\ B(p, q) &= \max(|q|, |3p + q|, |3p + 2q|). \end{aligned} \tag{9}$$

The spans in (8) are respectively A, B, A, B. Here and below $t_+ = \max(0, t)$.

Proposition 4. *For all integers $n \geq 0$,*

$$a(n) = \sum_{(p,q) \in \mathbb{Z}^2} (n + 1 - A(p, q))_+^2 (n + 1 - B(p, q))_+^2. \tag{10}$$

The sum is finite; only $-n \leq p, q \leq n$ can contribute.

Proof. The four bases are independent once p, q are fixed, so multiply their four interval counts. Lemma 3 ensures a bijection. Since $A \geq |p|$ and $B \geq |q|$, the asserted finite range follows. □

3 A symmetry of the counting sum

Use the unimodular change of variables $x = p$, $y = p + q$. Then

$$A = \max(|x|, |y|, |x + y|), \qquad B = \max(|y - x|, |2x + y|, |x + 2y|).$$

Both are invariant under the integral maps

$$R(x, y) = (-y, x + y), \qquad S(x, y) = (y, x).$$

Indeed each map permutes, up to signs, the three forms in each maximum. Direct multiplication gives $R^6 = S^2 = 1$ and $SRS = R^{-1}$.

Lemma 5. *Each orbit of this group on $\mathbb{R}^2$ has exactly one representative in the closed chamber $x \geq y \geq 0$. Write it as $(x, y) = (h + k, h)$, with $h, k \geq 0$. The orbit size is*

$$\omega(h,k) = \begin{cases} 12, & h > 0,\ k > 0, \\ 6, & \text{exactly one of } h, k \text{ is positive}, \\ 1, & h = k = 0. \end{cases} \tag{11}$$

For this representative, $A = 2h + k$ and $B = 3h + 2k$.

Proof. The successive rays $R^j(1, 0)$, $0 \leq j < 6$, are

$$(1,0), (0,1), (-1,1), (-1,0), (0,-1), (1,-1).$$

The six images under R of the first quadrant are the sectors between successive rays. They cover the plane with disjoint interiors. The map S interchanges the two halves of the first quadrant, separated by $x = y$. This gives the twelve chamber interiors. Their images contain twelve distinct points for a point with $x > y > 0$. Each nonzero boundary ray is fixed pointwise by its reflecting involution, and only the two adjacent closed chamber images meet there. Its nonzero points therefore have orbit size six. The origin is fixed. Within the closed chosen chamber no two distinct points can therefore belong to the same orbit. Evaluating the two maxima there proves the last assertion. □

It follows from Proposition 4 that

$$a(n) = \sum_{\substack{h,k \geq 0 \\ 3h+2k \leq n}} \omega(h,k)(n + 1 - 2h - k)^2(n + 1 - 3h - 2k)^2, \tag{12}$$

where h, k are integers. The weights restore all lattice points; this use of symmetry does *not* identify or quotient labeled arrays. Boundary weights 6 and the origin weight 1 are essential.

For example, when $n = 1$ only $(h, k) = (0, 0)$ contributes, giving 16. For $n = 2$ the origin contributes 81 and $(0, 1)$ contributes $6 \cdot 2^2 \cdot 1^2 = 24$, giving 105. For $n = 3$ the contributions at $(0, 0), (0, 1), (1, 0)$ are $256, 216, 24$, giving 496. Thus the index convention follows directly from the counting argument.

4 Deriving and extracting the generating function

Put $r = n - 3h - 2k$ in (12). As a formal power series,

$$F(z) = \sum_{h,k,r \geq 0} \omega(h,k)(r + 1)^2(r + h + k + 1)^2 z^{3h+2k+r}. \tag{13}$$

For each coefficient only finitely many triples occur, so rearranging these sums is justified without an analytic convergence assumption. Let $w_0 = 1$ and $w_j = 2$ for $j > 0$. Then

$$\omega(h,k) = 3w_h w_k - 2\,\mathbf{1}_{h=0}\mathbf{1}_{k=0}.$$

Define

$$T_0(t) = \frac{1+t}{1-t}, \qquad T_1(t) = \frac{2t}{(1-t)^2}, \qquad T_2(t) = \frac{2t(1+t)}{(1-t)^3},$$

$$U_2(z) = \frac{1+z}{(1-z)^3}, \qquad U_3(z) = \frac{1+4z+z^2}{(1-z)^4}, \qquad U_4(z) = \frac{1+11z+11z^2+z^3}{(1-z)^5}.$$

Here $T_j(t) = \sum_{\ell \geq 0} w_\ell \ell^j t^\ell$ for $j = 1, 2$, $T_0(t) = \sum_{\ell \geq 0} w_\ell t^\ell$, and $U_j(z) = \sum_{r \geq 0} (r+1)^j z^r$. These identities follow by differentiating the geometric series. Set $P_j = T_j(z^3)$ and $Q_j = T_j(z^2)$. Expanding $(r+h+k+1)^2$ in (13) gives

$$\begin{aligned} F = 3\big[&(P_2Q_0 + 2P_1Q_1 + P_0Q_2)U_2 \\ &+ 2(P_1Q_0 + P_0Q_1)U_3 + P_0Q_0U_4\big] - 2U_4. \end{aligned} \tag{14}$$

Substitution and collecting terms yields (3), proving Theorem 2. This identity is also checked by exact polynomial cross multiplication in the supplement.

For completeness, the following coefficient extraction makes all six constituents explicit without interpolation. Define

$$P(n) = \frac{780n^6 + 4680n^5 + 13650n^4 + 23400n^3 + 33936n^2 + 33552n + 10745}{7776}. \tag{15}$$

Then

$$a(n) = P(n) - \frac{3}{32}(-1)^n + C_{n \bmod 3}(n), \tag{16}$$

where

$$\begin{aligned} C_0(n) &= \frac{2(3n^2 - 18n - 35)}{243}, \\ C_1(n) &= \frac{2(3n^2 + 30n + 13)}{243}, \\ C_2(n) &= -\frac{4(3n^2 + 6n - 11)}{243}. \end{aligned} \tag{17}$$

Here is a verifiable rational-algebra certificate for (16). Let

$$(b_1, \ldots, b_7) = \left(-\frac{2521}{7776}, -\frac{737}{432}, \frac{2687}{216}, -\frac{2275}{36}, \frac{325}{2}, -\frac{1625}{9}, \frac{650}{9}\right)$$

and

$$H(z) = -70 + 92z - 52z^2 + 86z^3 + 86z^4 - 220z^5 + 92z^6 - 70z^7 + 56z^8.$$

Multiplying denominators verifies

$$F(z) = \sum_{j=1}^{7} \frac{b_j}{(1-z)^j} - \frac{3}{32(1+z)} + \frac{H(z)}{243(1-z^3)^3}. \tag{18}$$

The first sum has coefficient $\sum_{j=1}^{7} b_j \binom{n+j-1}{j-1} = P(n)$. The second term has coefficient $-3(-1)^n/32$. If $n = 3k + s$, $0 \leq s < 3$, and $H(z) = \sum_i H_i z^i$, the last term has coefficient

$$\frac{1}{243} \sum_{j=0}^{2} H_{s+3j} \binom{k-j+2}{2}. \tag{19}$$

The binomial is interpreted as the polynomial $v(v-1)/2$ at $v = k - j + 2$. When $j > k$ the only possible values here are $v = 0, 1$, and it vanishes, so this formula holds even for $k = 0, 1$. Expanding (19) gives exactly $C_s(3k + s)$. Finally substitute $n = 6m + r$ in (16) and collect powers of m. This gives Table 1 and (2), proving all six formulas for every $m \geq 0$.

Proof of the degree and minimal period assertions. Equation (16) gives a quasipolynomial of degree six with period dividing six. The rational function (3) has $N(1) = 3900$, $N(-1) = -12$, and remainder 12 on division of $N(z)$ by $z^2 + z + 1$. Thus it has a pole of order seven at 1, a pole at -1, and poles at both primitive cube roots. If a sequence is a quasipolynomial with period d, summing its constituent polynomials over residue classes shows that all poles of its generating function are dth roots of unity. The poles at -1 and primitive cube roots force $2 \mid d$ and $3 \mid d$. Hence the least period is six. The order-seven pole at 1 also confirms degree six. □

5 An independent lattice-polytope period bound

Let $\mathcal{P} \subset \mathbb{R}^6$ be defined by requiring each expression in (6) to lie in $[0,1]$. Lemma 3 identifies the arrays exactly with $n\mathcal{P} \cap \mathbb{Z}^6$, including $n = 0$. The polytope is bounded: $a, c, e, f \in [0,1]$, and p, q are differences of entries in that interval. The point

$$(a, c, e, f, p, q) = (1/2, 1/2, 1/2, 1/2, 0, 0)$$

makes all twelve expressions strictly between 0 and 1. Consequently $\mathcal{P}$ has dimension six.

Proposition 6. *Every vertex of $\mathcal{P}$ has all coordinates with a common denominator dividing one of $1, 2, 3$. In particular $6\mathcal{P}$ is a lattice polytope.*

Proof. At a vertex, at least one of the three expressions in each base group of (8) must be at a bound. Otherwise that base coordinate can move in both directions, contradicting extremality. Choose one tight row in each of the four groups. Subtract its equation from all other tight equations in that group. These differences involve only p, q. Since the complete active system at a vertex has rank six, the difference equations have rank two. Choose two independent ones. Each has integral right side and left side equal, up to sign, to one of

$$p,\ \ q,\ \ p+q,\ \ 2p+q,\ \ 3p+q,\ \ 3p+2q.$$

For the six coefficient vectors $(1,0), (0,1), (1,1), (2,1), (3,1), (3,2)$, the fifteen pairwise absolute determinants, in lexicographic pair order, are

$$1, 1, 1, 1, 2, \quad 1, 2, 3, 3, \quad 1, 2, 1, \quad 1, 1, \quad 3.$$

Thus the chosen determinant has absolute value 1, 2, or 3. Cramer's rule gives the same denominator bound for p, q. Each chosen base equation then determines its base as an integer minus an integral linear combination of p, q. This proves the claim. □

The rational form of Ehrhart's theorem [3, Chapter 3] now gives independently a degree-six quasipolynomial whose period divides six. All inequalities were closed and the lattice is exactly $\mathbb{Z}^6$, so no boundary term or lattice-index adjustment is omitted. This gives a second proof route: seven exact values in each residue class determine all six constituents. The 42 values in Table 2 supply such a certificate. The direct generating function proof above does not need this finite interpolation route.

6 Exact verification and reproducibility

The supplemental code uses only Python's standard library and exact integers or rational numbers. Its roles are distinct:

- The structural check builds the original line-sum equations, checks the integral parameter map and inverse, and verifies the fifteen determinants. As a redundant check it computes all $\binom{12}{6} = 924$ maximal minors of the parameter matrix; the absolute-determinant multiplicities are $0 : 384$, $1 : 324$, $2 : 108$, $3 : 108$.
- The counting check evaluates both (10) and (12) for $0 \le n \le 41$ and compares the results with the six formulas and the saved certificate. It also exhaustively enumerates the original twelve entries for $0 \le n \le 3$, testing (1) directly, and obtains $1, 16, 105, 496$.
- The algebra check proves equality of rational functions by polynomial cross multiplication: (14), (3), (18), and the generating function of all six polynomials are compared exactly. This verifies identities, not only a finite Taylor truncation.

- A separate source audit compares all six printed source constituents as exact rational polynomials and verifies the shift and coefficient correction in (4).

The verification command, from the package root, is

```
python src/verify.py
```

It requires Python 3.9 or later and no third-party packages. It can also be invoked by its path from another working directory. To regenerate the finite count certificate, run

```
python src/counting.py --write
```

To rebuild the paper, run from the package root:

```
pdflatex -interaction=nonstopmode -halt-on-error \
  -jobname=research_paper -output-directory=output paper/main.tex
```

Run the LaTeX command twice to resolve cross-references. The backslash denotes a shell line continuation; on Windows PowerShell, put the command on one line instead. The supplement contains the full sources, the count certificate, the literature audit, and the proof map.

m	$a(6m)$	$a(6m+1)$	$a(6m+2)$	$a(6m+3)$	$a(6m+4)$	$a(6m+5)$
0	1	16	105	496	1759	5052
1	12469	27412	55059	102952	181543	304908
2	491563	765184	1155567	1699684	2442553	3438468
3	4752283	6460432	8652429	11432392	14920189	19253232
4	24588229	31102456	38995845	48492976	59844451	73329300
5	89257201	107969956	129844215	155293936	184771939	218772972
6	257836399	302547832	353542539	411508372	477187453	551379948

Table 2: Exact counts covering seven points in each residue class.

Scope and verification status

All six conjectured formulas are proved for the full domain $n \geq 0$, with labeled positions and exactly the stated line-sum conditions. The proof is elementary apart from the optional independent Ehrhart argument. Exact computations audit the arithmetic and transcription. The manuscript and code were developed by GPT-6 Astra, an AI system, including separate internal checks. No independent human peer review or Lean formal verification is claimed. The formulas retain their original attribution; a claim of worldwide priority is not made.

References

[1] R. H. Hardin, *OEIS A195806: Number of triangular* $0, \ldots, n$ *arrays with five rows*, OEIS Foundation. Formulas contributed by M. Kauers and C. Koutschan, 1 March 2023. https://oeis.org/A195806. Accessed 28 October 2026.

[2] M. Kauers and C. Koutschan, *Some D-Finite and Some Possibly D-Finite Sequences in the OEIS*, Journal of Integer Sequences **26** (2023), Article 23.4.5, especially §5.1, Conjecture 11 and Theorem 13. https://cs.uwaterloo.ca/journals/JIS/VOL26/Koutschan/kout4.pdf. Preprint: https://arxiv.org/abs/2303.02793, v2, 24 April 2023. Computational supplement: https://www.koutschan.de/data/seq/SequencesOEIS.pdf.

[3] M. Beck and S. Robins, *Computing the Continuous Discretely: Integer-Point Enumeration in Polyhedra*, 2nd ed., Springer, 2015, Chapter 3. https://doi.org/10.1007/978-1-4939-2969-6.